

Reinforcement Learning for Optimal Setpoint Determination in Energy Utility Systems

Master Thesis

Master of Science in Artificial Intelligence

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Task description

Energy utility systems is the backbone of modern energy infrastructures. Their complexity ranges from small, localized systems to vast, highly interconnected infrastructure networks. Most of them operate under dynamic conditions where multiple interacting variables influence operational efficiency, cost, reliability, and stability. Conventionally, optimal setpoints for such systems are determined using mathematical optimisation. These approaches rely on accurate system models and are typically formulated as constrained optimisation problems. While effective for well-defined and moderately sized systems, classical optimisation becomes computationally expensive as system complexity increases. Furthermore, time-varying dynamics, uncertainty from renewable source integration, and model-plant mismatches challenge the applicability of purely model-based approaches.

The central problem addressed in this thesis is whether reinforcement learning can reliably and efficiently determine optimal setpoints for two distinct systems: a simple battery energy storage system and a more complex and non-linear cookie production factory. The challenge arises from the need to simultaneously minimise operational costs, maintain constraint satisfaction, and ensure stable system behaviour under uncertainty. Classical optimisation methods struggle with scalability and adaptability, particularly in dynamic environments where conditions change continuously. Reinforcement Learning (RL) has emerged as a promising data-driven paradigm that learns optimal decision policies through interaction with the environment, without requiring explicit analytical system models. Despite its success in robotics and continuous benchmarks, the application of RL to complex energy utility systems remains insufficiently explored. This thesis proposes to formulate the setpoint optimisation problem as a Markov Decision Process (MDP) and to develop a reinforcement learning framework capable of learning adaptive policies. By comparing RL-based approaches with established optimisation methods, the research aims to identify strengths, weaknesses, and application boundaries of both approaches.

The evaluation of the proposed method will be conducted through systematic simulation studies on two energy system case studies of varying complexity. Performance will be assessed using quantitative metrics including total operational cost, constraint violation frequency, and computational time. The results will provide a structured comparison between reinforcement learning and classical optimisation and will determine under which conditions RL offers practical advantages for real-time setpoint optimisation in complex energy utility systems.

Abstract

Example summary in English.

Zusammenfassung

Beispielzusammenfassung in Deutsch.

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1 Introduction

Introduces the research problem, motivation, and objectives of the thesis.

2 Theory and Background

Provides the fundamental concepts and background required to understand the systems, optimization methods, and reinforcement learning used in this work.

Optimization is the process of selecting the best decision or solution, based on a defined objective, from a set of feasible alternatives while respecting given constraints. In energy systems, this typically involves determining control actions that minimize operational costs or energy consumption under physical and operational constraints.

Mathematically, optimization is formulated as:

$$\min_{x \in \Omega} f(x) \quad \text{s.t.} \quad h(x) = 0, \quad g(x) \leq 0, \quad (1)$$

where $x \in \Omega \subseteq \mathbb{R}^n$ is the decision variable, $f : \mathbb{R}^n \rightarrow \mathbb{R}$ the objective function, $h : \mathbb{R}^n \rightarrow \mathbb{R}^p$ the equality constraints, and $g : \mathbb{R}^n \rightarrow \mathbb{R}^q$ the inequality constraints. The goal is to find the optimal solution x^* that achieves the minimum or maximum value $f(x^*)$.

In practice, optimization problems can be addressed using different methods depending on the availability of accurate system models, the presence of uncertainty, and the complexity of decision-making.

In this thesis, two approaches are considered in the context of energy systems:

- Mathematical optimization, which relies on explicit system models and deterministic formulations, and is used as a baseline for performance comparison;
- Reinforcement learning, which learns decision-making policies through interaction with the environment and constitutes the main approach of this work.

The following sections introduce the fundamental principles of both approaches.

2.1 Mathematical Optimization

Presents traditional optimization methods used for determining optimal system set-points.

Mathematical optimization uses explicit models of system behavior, constraints, and objectives to solve decision problems. In energy systems, it is commonly used to compute optimal operating strategies by minimizing costs or maximizing efficiency while respecting physical and operational limits. In this thesis, mathematical optimization is used as a baseline method for performance comparison, rather than being further developed as the main approach.

An optimization problem is typically formulated using a set of decision variables, an objective function, and a set of constraints. The decision variables represent the control actions to be determined, such as the charging and discharging of a battery or production levels in a factory. The objective function defines the performance criterion to be optimized, commonly expressed as cost minimization or profit maximization. Constraints capture the physical and operational limits of the system, including capacity bounds, energy balance equations, and technical restrictions.

Depending on the problem structure, optimization problems can be classified as linear, nonlinear, or mixed-integer. In energy system applications, mixed-integer linear programming (MILP) is widely used, as it enables the modeling of both continuous and discrete decision variables while maintaining computational tractability.

However, as system size and complexity increase, such formulations can become computationally demanding and require accurate system modeling.

A key distinction in optimization is between deterministic and stochastic formulations. In deterministic optimization, all parameters, such as electricity prices or demand, are assumed to be known in advance. In contrast, stochastic optimization explicitly accounts for uncertainty by modeling uncertain parameters as random variables. While stochastic approaches provide a more realistic representation, they often lead to increased computational complexity.

Energy system optimization problems are often time-coupled, meaning that decisions made at one time step influence future system states. For example, the state of charge of a battery depends on previous charging and discharging actions, which introduces inter-temporal dependencies into the optimization problem.

These problems are typically solved using dedicated optimization solvers, such as those designed for linear and mixed-integer programming which can efficiently compute optimal solutions for well-defined models. However, their performance depends strongly on the accuracy of system models and assumptions, and solution times can increase significantly for large-scale or highly dynamic problems, limiting their applicability in real-time decision-making.

These limitations motivate the use of reinforcement learning, which can learn control strategies directly from interaction with the environment without requiring explicit system models, aiming to minimize operational costs or maximize economic profit, and therefore constitutes the main approach of this thesis.

2.2 Reinforcement Learning

[Introduces reinforcement learning as a data-driven approach for learning optimal control policies through interaction with the environment.](#)

Reinforcement learning (RL) is a data-driven approach to sequential decision-making, in which an agent learns to take optimal actions through interaction with an environment. Unlike mathematical optimization, RL does not require an explicit model of system dynamics. Instead, it learns control strategies based on observed state transitions and received rewards, making it well-suited for complex and uncertain environments.

In the context of energy systems, reinforcement learning can be used to determine control actions, such as battery charging and discharging, in order to minimize operational costs or maximize economic profit over time. By continuously interacting with the system, the agent improves its decision-making policy based on experience.

Formally, reinforcement learning problems are typically modeled as a Markov Decision Process (MDP), defined by the tuple

$$(\mathcal{S}, \mathcal{A}, P, R, \gamma),$$

where:

- $\mathcal{S}$ is the set of states representing the system,
- $\mathcal{A}$ is the set of actions,
- $P(s' | s, a)$ defines the transition dynamics,
- $R(s, a)$ is the reward function,
- $\gamma \in [0, 1]$ is the discount factor.

At each time step, the agent observes the current state s , selects an action a , and transitions to a new state s' while receiving a reward. The objective is to learn a policy $\pi(a | s)$ that maximizes the expected cumulative reward over time.

In energy system applications, the underlying decision problem can often be formulated as a stochastic optimal control problem, where uncertainties arise from exogenous variables such as renewable generation and electricity prices. These uncertainties are commonly modeled as stochastic processes, and the resulting problem can be expressed as a finite-horizon MDP [1]. The goal is typically to minimize expected operational costs over time while respecting system dynamics and constraints.

Classical approaches such as dynamic programming can be used to solve such MDPs through backward recursion. However, these methods suffer from the curse of dimensionality when the state space becomes large [1]. This limitation makes exact solutions computationally intractable for realistic energy systems.

Reinforcement learning provides an alternative by learning approximate solutions directly from data. In particular, methods such as Q-learning can approximate the optimal value function without requiring full knowledge of the transition probabilities. As highlighted in [1], RL methods are especially useful in high-dimensional settings where traditional optimization methods become infeasible.

Furthermore, advanced RL methods such as policy gradient and actor-critic algorithms are better suited for problems with continuous or high-dimensional action spaces, as they avoid exhaustive discretization of actions and instead learn parameterized policies [1]. This is particularly important for energy systems, where control decisions are naturally continuous.

Despite these advantages, reinforcement learning introduces new challenges, including the need for sufficient training data, potential instability during learning, and sensitivity to reward design and state representation.

In this thesis, reinforcement learning is used as the primary approach to learn control policies for energy system operation under uncertainty. Its performance is evaluated against a mathematical optimization benchmark, highlighting the trade-offs between model-based and data-driven approaches.

3 Use Cases

Describes the two systems considered in this thesis and their operational challenges.

3.1 Battery Energy Storage System

Describes the battery system used as the first case study.

3.2 Cookie Production Factory

Introduces the industrial production system considered as the second case study.

4 Methodology

Describes how the optimization problem is formulated as a reinforcement learning problem.

4.1 Problem Formulation

Formulates the system control problem as a reinforcement learning problem.

4.2 Reinforcement Learning Framework

Explains the structure of the RL agent and learning algorithm.

4.3 Training Setup

Describes the training environment, reward function, and hyperparameter configuration.

4.4 Evaluation Metrics

Defines the metrics used to evaluate the performance of the RL agent and optimization baseline.

5 Results and Discussion

Presents and analyzes the experimental results.

5.1 Battery Energy Storage System

5.2 Cookie Production Factory

5.3 Key Findings

6 Conclusion and Future Work

Summarizes the main findings and outlines possible future work.